

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460951.6799 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.025
Transit depth (Rp/Rs)^2	2.39 +/- 0.78 [%]
Orbital Inclination (inc)	89.61 +/- 1.56
Airmass coefficient 1 (a1)	1.332 +/- 0.035
Airmass coefficient 2 (a2)	-0.178 +/- 0.017
Scatter in the residuals of the lightcurve fit is	2.64 %
Best Comparison Star	None
Optimal Aperture	3.25
Optimal Annulus	15.42
Transit Duration (day)	0.1037 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.025 vs 0.1394 (deviation 0.49σ)
Duration fit vs published	0.1037 d vs 0.0983 d (deviation 1.0σ)
Mid-time O–C	-12.6 min
Depth SNR	4.9
Residual scatter	2.64 %

Light curve

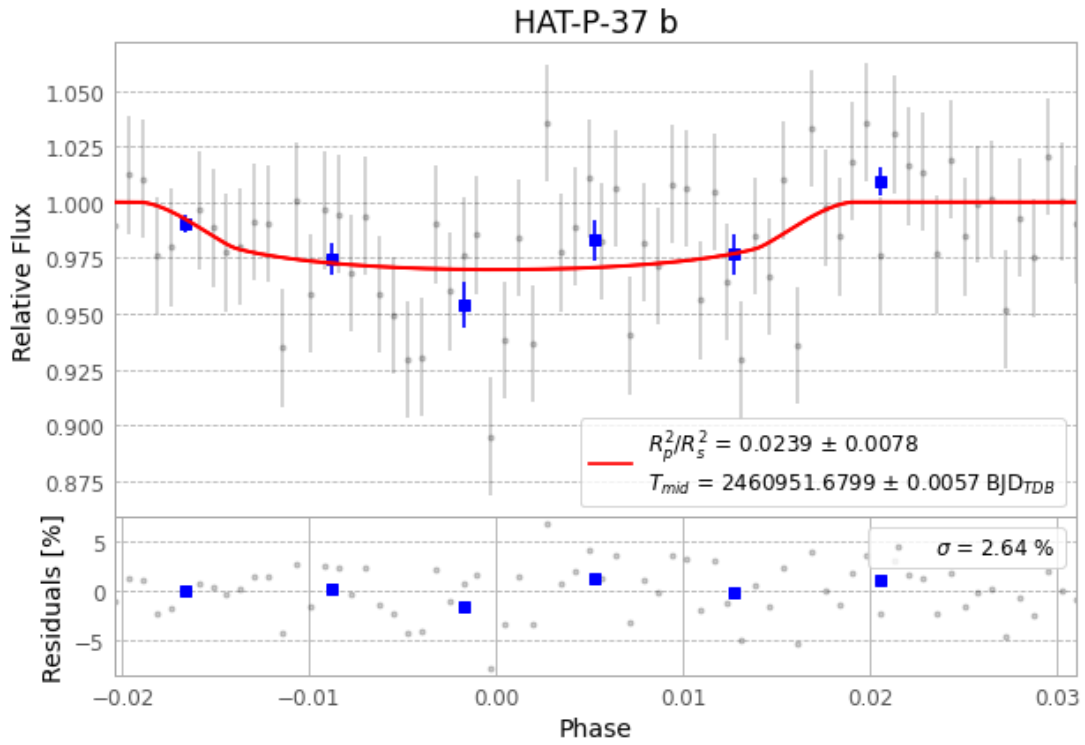


Figure 1 — FinalLightCurve_HAT-P-37 b_2025-10-02.png (SHA-256 15180788a6aa53d4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 334], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 225784d51bee7333...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), HATP-37251003025415.FITS → HATP-37251003062116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-251003.log (b23dc2f86cb3...), FinalLightCurve_HAT-P-37 b_2025-10-02.png (15180788a6aa...), FinalLightCurve_HAT-P-37 b_2025-10-02.csv (74e619d07452...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:45Z.